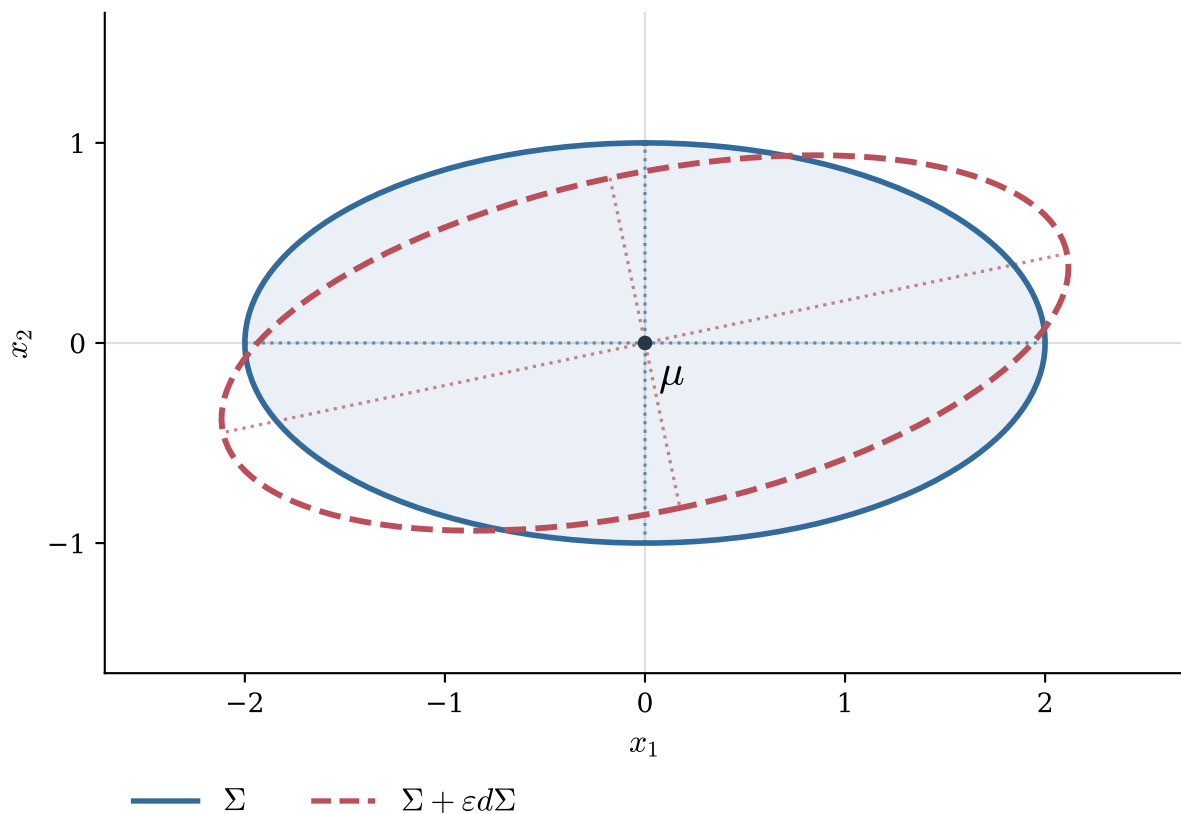


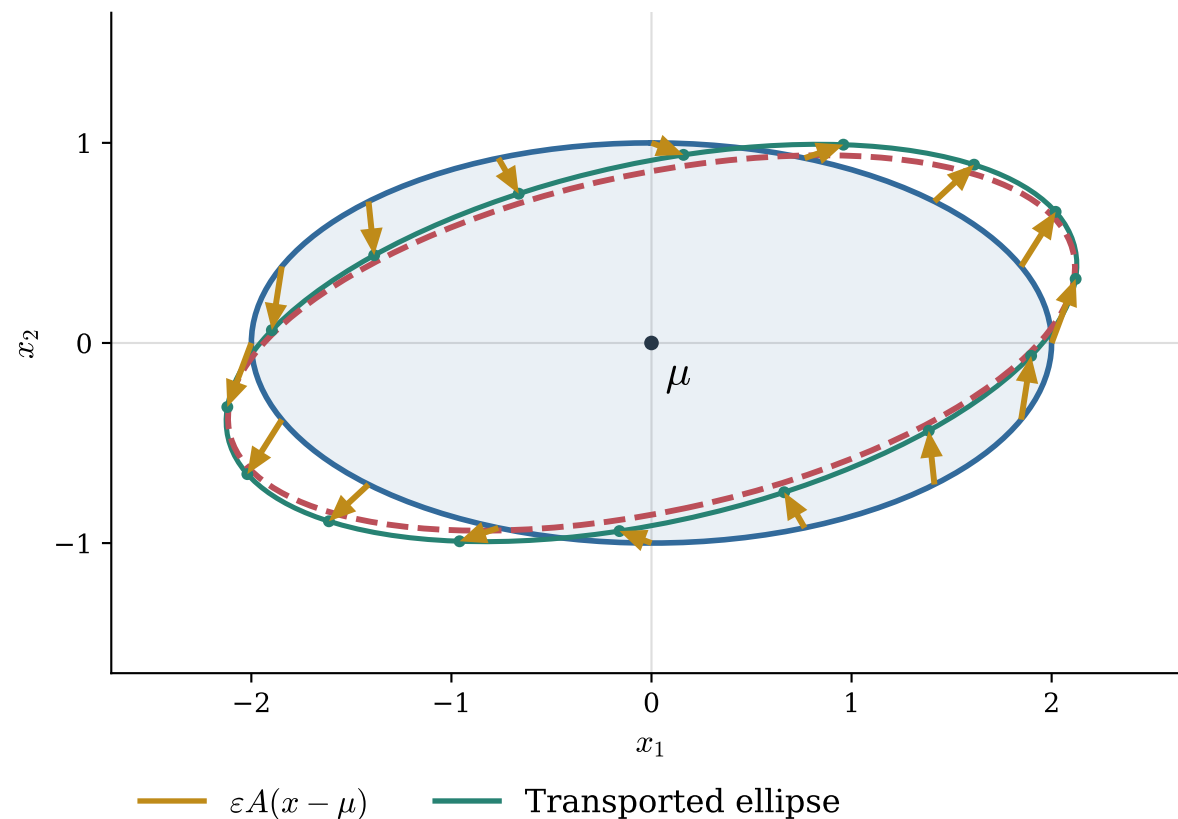
From covariance variation to a velocity field

$$d\Sigma \longmapsto A = \mathcal{L}_\Sigma(d\Sigma), \quad \Sigma A + A \Sigma = d\Sigma$$

(a) Prescribe a covariance change



(b) Recover the gradient velocity



$$(I + \varepsilon A)\Sigma(I + \varepsilon A)^T = \Sigma + \varepsilon d\Sigma + \varepsilon^2 A\Sigma A$$

The prescribed change and the transported covariance agree to first order.

In the principal basis: $(R^T A R)_{ij} = \frac{(R^T d\Sigma R)_{ij}}{\sigma_i^2 + \sigma_j^2} \quad (\varepsilon = 0.4 \text{ for display})$